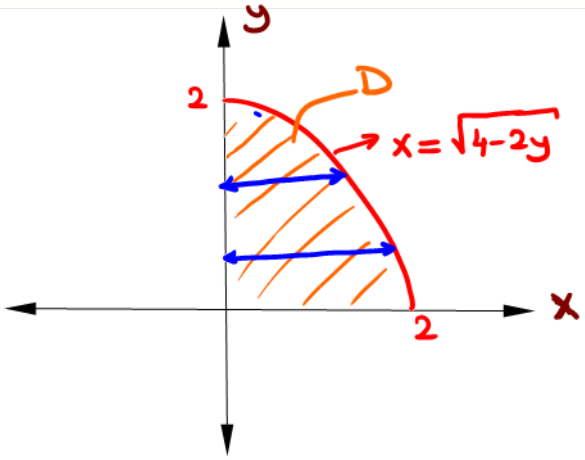
~ MAT1072 Matematik 2 Test Soruları - 2 ~

~ Çözümler ~

1. D birinci çeyrek bölgede $x^2 = 4-2y$ eğrisi ile sınırlı kapalı bir bölge olduğuna göre bu bölge üzerinde $\iint_D \frac{1}{\sqrt{2y-y^2}}\,dA$ integralinin değeri aşağıdakilerden hangisidir?

a) 1 b) 4 c) -4 d) -1 e) $\frac{3}{2}$



$$\iint_D \frac{1}{\sqrt{2y-y^2}}\,dA = \int_{y=0}^{y=2}\int_{0}^{\sqrt{4-2y}} \frac{1}{\sqrt{y}\sqrt{2-y}}\,dx\,dy$$

$y(2-y)$

$$= \int_0^2 x\Big|_0^{\sqrt{4-2y}} \frac{1}{\sqrt{y}\sqrt{2-y}}\,dy$$

$$= \int_0^2 \sqrt{2}\cdot\cancel{\sqrt{2-y}}\,\frac{1}{\sqrt{y}\cdot\cancel{\sqrt{2-y}}}\,dy = \int_0^2 \sqrt{2}\,\frac{dy}{\sqrt{y}}$$

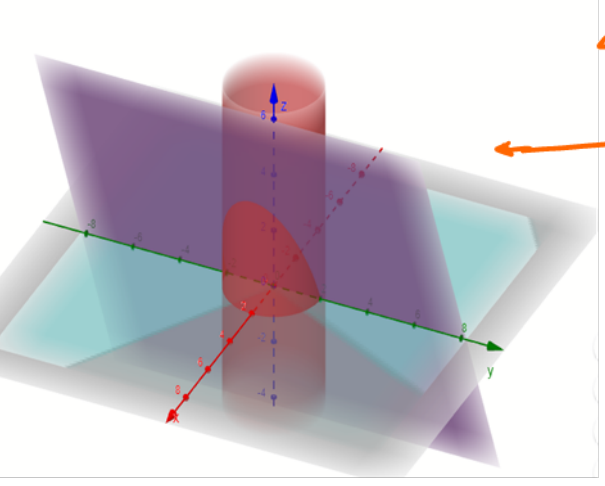
$$= \sqrt{2}\cdot 2\sqrt{y}\Big|_0^2 = 2\sqrt{2}\cdot\sqrt{2} = 4$$

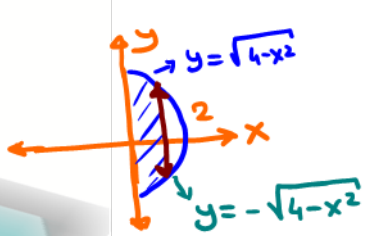
B şıkkı

2. $x^2+y^2 \le 4$ ve $0 \le z \le 2x$ eşitsizlikleri ile belirlenen bölgenin hacmini veren katlı integral aşağıdakilerden hangisidir?

a) $\int_0^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} 2x\,dy\,dx$ b) $\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} 2x\,dy\,dx$ c) $\int_0^2 \int_0^{\sqrt{4-x^2}} 2x\,dy\,dx$

d) $\int_0^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} 2x\,dy\,dx$ e) $\int_{-2}^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} 2x\,dx\,dy$





$$x^2+y^2 \le 4 \quad \Big] \; -\sqrt{4-x^2} \le y \le \sqrt{4-x^2}$$
$$0 \le z \le 2x \quad \Big] \; 0 \le 2x \le 4$$
$$0 \le x \le 2$$

$$V = \int_{x=0}^{x=2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (2x-0)\,dy\,dx$$

D şıkkı

3. $\int_0^{\frac{1}{\sqrt{2}}} \int_x^{\sqrt{1-x^2}} \sqrt{x^2+y^2}\, dydx$ integralinin değeri aşağıdakilerden hangisidir?

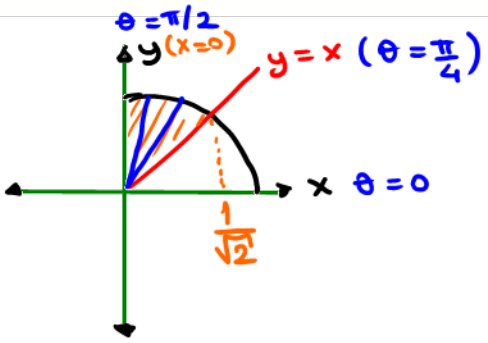
a) $\pi - e$ b) $-\frac{\pi}{6}$ c) $\frac{\pi}{6}$ d) $-\frac{\pi}{12}$ e) $\frac{\pi}{12}$

$x=0 \quad y=x$

$x=\frac{1}{\sqrt{2}} \quad y=\sqrt{1-x^2}$

$x^2+y^2=r^2$

$dydx = r\,dr\,d\theta$



$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \int_0^1 r \cdot r\, dr\, d\theta$$

$$= \int_{\frac{\pi}{4}}^{\pi/2} \frac{r^3}{3}\Big|_0^1 d\theta = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1}{3} d\theta$$

$$= \frac{1}{3}\theta\Big|_{\frac{\pi}{4}}^{\pi/2} = \frac{1}{3}\left(\frac{\pi}{2}-\frac{\pi}{4}\right)$$

$$= \frac{\pi}{12}$$

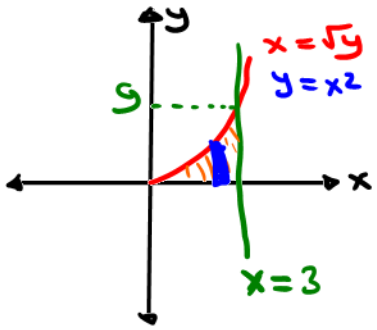
E şıkkı

4. $\int_0^9 \int_{\sqrt{y}}^3 e^{x^3}\, dxdy = ?$ → integral bu haliyle çözülemez! Sıralama değişmeli!

a) $\frac{1}{3}(e^9-1)$ b) $\frac{1}{2}(e^{27}-1)$ c) $\frac{1}{3}(e^{27}-1)$ d) $\frac{1}{3}(e^3-1)$ e) $\frac{1}{3}(e^{27}+1)$

$y=0 \quad x=\sqrt{y}$

$y=9 \quad x=3$



$dxdy \rightarrow dydx$ olmalı!

$$\int_0^3 \int_0^{x^2} e^{x^3}\, dydx$$

$$= \int_0^3 y\Big|_0^{x^2} e^{x^3} dx = \int_0^3 x^2 e^{x^3} dx$$

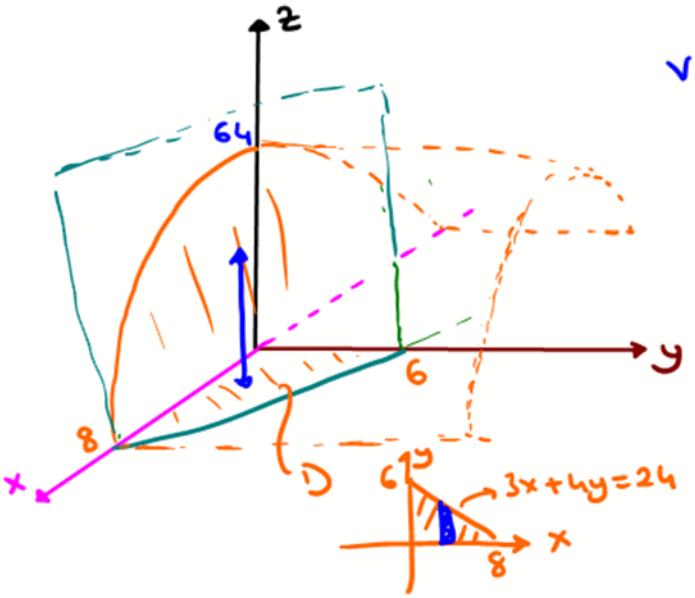
$$= \frac{1}{3} e^{x^3}\Big|_0^3 = \frac{1}{3}(e^{27}-1)$$

C şıkkı

5. Birinci sekizde birlik bölgede $x^2+z=64$ yüzeyi ile $x=0$, $y=0$, $z=0$ ve $3x+4y=24$ düzlemleri ile sınırlı cismin hacmini veren integral aşağıdakilerden hangisidir?

a) $\int_0^8 \int_0^{(6-\frac{3}{4}x)} (64-x^2)\,dydx$ b) $\int_0^6 \int_0^{(6-\frac{3}{4}x)} (64-x^2)\,dydx$ c) $\int_0^8 \int_0^{(8-\frac{4}{3}x)} (64-x^2)\,dydx$

d) $\int_0^9 \int_0^{(6-\frac{3}{4}x)} (64-x^2)\,dydx$ e) $\int_0^7 \int_0^{(6-\frac{3}{4}x)} (64-x^2)\,dydx$



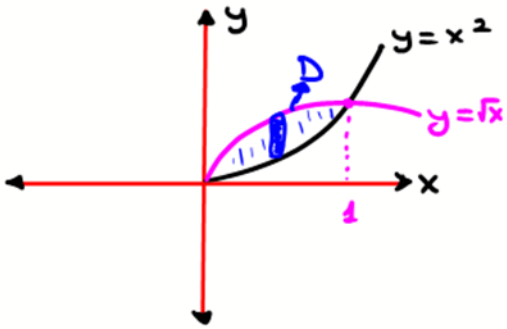
$$V=\iint_D (64-x^2-0)\,dA$$

$$V=\int_0^8 \int_0^{6-\frac{3}{4}x} (64-x^2)\,dydx$$

A şıkkı

6. $f(x,y)=\frac{1}{x}$ fonksiyonunun $0\le x\le 1$, $x^2\le y\le \sqrt{x}$ bölgesi üzerindeki ortalama değeri aşağıdakilerden hangisidir?

a) $\frac{9}{4}$ b) $\frac{9}{2}$ c) $\frac{9}{5}$ d) $\frac{3}{4}$ e) $\frac{3}{2}$



$$ort f=\overline{f}=\frac{1}{D\text{'nin alanı}}\iint_D f(x,y)\,dA=\frac{\text{Hacim}}{\text{Alan}}$$

$$\text{Alan}=\int_0^1 \int_{x^2}^{\sqrt{x}} dydx=\int_0^1 (\sqrt{x}-x^2)\,dx=\frac{2}{3}x^{3/2}-\frac{x^3}{3}\Big|_0^1$$

$$=\frac{2}{3}-\frac{1}{3}=\frac{1}{3}$$

$$\text{Hacim}=\int_0^1 \int_{x^2}^{\sqrt{x}} \frac{1}{x}\,dydx=\int_0^1 \left(\frac{1}{\sqrt{x}}-x\right)dx=\lim_{R\to 0^+}\int_R^1 \left(\frac{1}{\sqrt{x}}-x\right)dx$$

$$=\lim_{R\to 0^+}\left(2\sqrt{x}-\frac{x^2}{2}\right)\Big|_R^1=\lim_{R\to 0^+}\left(2-\frac{1}{2}-2\sqrt{R}+\frac{R^2}{2}\right)=\frac{3}{2}$$

$$ort f=\frac{\text{Hacim}}{\text{Alan}}=\frac{3/2}{1/3}=\frac{9}{2}$$

B şıkkı

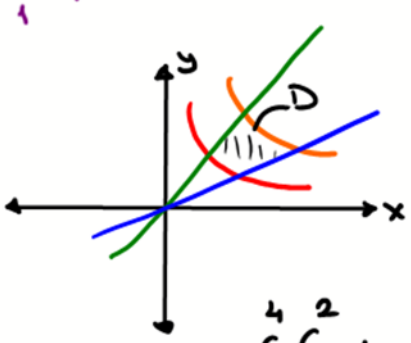
7. Birinci bölgede $xy=1$, $xy=4$, $y=x$ ve $y=2x$ eğrileriyle sınırlı bölgenin alanını veren integral aşağıdakilerden hangisidir?

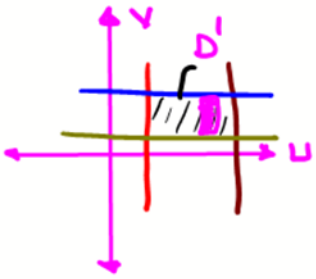
a) $\frac{1}{2}\int_1^2\int_1^4 \frac{1}{v}\,dudv$ b) $\frac{1}{4}\int_1^4\int_1^2 \frac{1}{v}\,dudv$ c) $\frac{1}{3}\int_1^4\int_1^2 \frac{1}{u}\,dudv$

d) $\frac{1}{2}\int_1^4\int_1^2 \frac{1}{v}\,dudv$ e) $\frac{1}{2}\int_1^4\int_1^2 \frac{1}{v}\,dvdu$

$xy=u$, $\frac{y}{x}=v$

D	D'
$xy=1$	$u=1$
$xy=4$	$u=4$
$\frac{y}{x}=1$	$v=1$
$\frac{y}{x}=2$	$v=2$





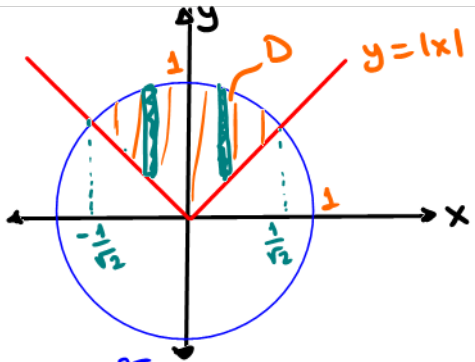
$$J=\frac{1}{\begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}}=\frac{1}{\begin{vmatrix} y & x \\ -y/x^2 & 1/x \end{vmatrix}}=\frac{1}{2\frac{y}{x}}=\frac{1}{2v}$$

$$\text{Alan}=\int_1^4\int_1^2 \frac{1}{2v}\,dvdu$$

E şıkkı

8. D, üstten $x^2+y^2=1$ eğrisi ve alttan $y=|x|$ ile sınırlı kapalı bir bölge olduğuna göre bu bölge üzerinde $\iint_D y\,dA$ integralinin değeri aşağıdakilerden hangisidir?

a) $\frac{\sqrt{2}}{3}-1$ b) $\frac{\sqrt{2}}{3}$ c) $-\frac{\sqrt{2}}{3}$ d) $\frac{2\sqrt{2}}{3}$ e) $\frac{\sqrt{2}}{2}$



2.yol $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\int_0^1 r\sin\theta\cdot r\,dr\,d\theta$

$=-\cos\theta\Big|_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\cdot\frac{r^3}{3}\Big|_0^1=\left(\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}\right)\cdot\frac{1}{3}=\sqrt{2}/3$

1.yol $\int_{-\frac{1}{\sqrt{2}}}^{0}\int_{-x}^{\sqrt{1-x^2}} y\,dy\,dx+\int_0^{\frac{1}{\sqrt{2}}}\int_x^{\sqrt{1-x^2}} y\,dy\,dx$

$=\int_{-\frac{1}{\sqrt{2}}}^{0}\frac{y^2}{2}\Big|_{-x}^{\sqrt{1-x^2}}dx+\int_0^{1/\sqrt{2}}\frac{y^2}{2}\Big|_x^{\sqrt{1-x^2}}dx$

$=\frac{1}{2}\int_{-\frac{1}{\sqrt{2}}}^{0}\left[(1-x^2)-x^2\right]dx+\frac{1}{2}\int_0^{1/\sqrt{2}}(1-x^2-x^2)\,dx$

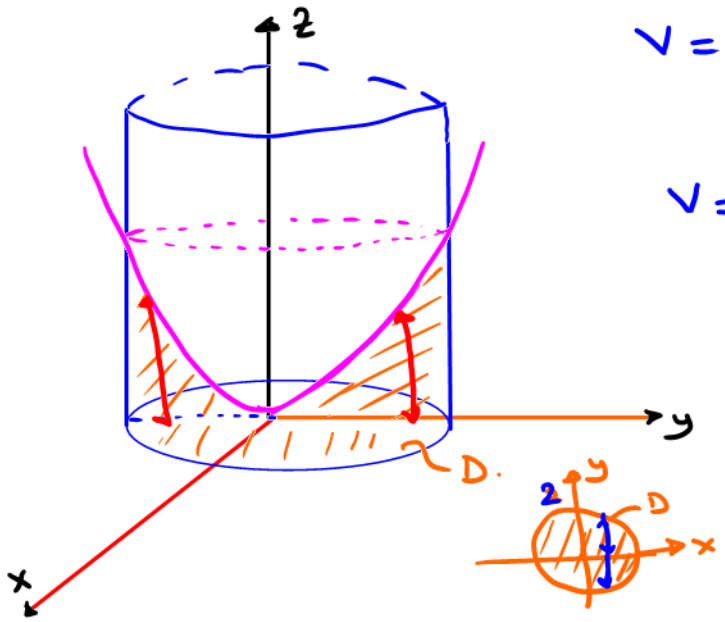
$=\frac{1}{2}\left(x-\frac{2}{3}x^3\right)\Big|_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}}=2\cdot\frac{1}{2}\left(\frac{1}{\sqrt{2}}-\frac{2}{3}\frac{1}{2\sqrt{2}}\right)=\frac{\sqrt{2}}{3}$

B şıkkı

9. $x^2+y^2=4$ silindirinin içinde $z=x^2+y^2$ paraboloidinin altında $z=0$ düzleminin üstünde oluşan cismin hacmini veren katlı integral aşağıdakilerden hangisidir?

a) $\int_0^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (x^2+y^2)\,dydx$ b) $\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (x^2+y^2)\,dydx$

c) $\int_0^2 \int_0^{\sqrt{4-y^2}} (x^2+y^2)\,dxdy$ d) $\int_0^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (x^2+y^2)\,dydx$ e) $\int_{-2}^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} (x^2+y^2)\,dydx$



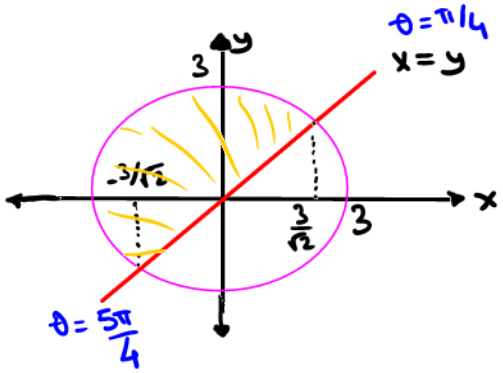
$$V=\iint_D (x^2+y^2-0)\,dA$$

$$V=\int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (x^2+y^2)\,dydx$$

B şıkkı

10. $$\int_{-\frac{3}{\sqrt{2}}}^{\frac{3}{\sqrt{2}}} \int_{-\sqrt{9-y^2}}^{y} yx^2\,dxdy + \int_{\frac{3}{\sqrt{2}}}^{3} \int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}} yx^2\,dxdy = ?$$

a) $\frac{81}{5\sqrt{2}}$ b) $\frac{81}{5}$ c) $\frac{81\sqrt{2}}{5}$ d) $\frac{81\sqrt{3}}{5}$ e) $\frac{162\sqrt{2}}{5}$



$$\int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} \int_0^3 r\sin\theta \cdot r^2\cos^2\theta \cdot r\,dr\,d\theta$$

$$= \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} \frac{r^5}{5}\Big|_0^3 \cdot \sin\theta\cdot\cos^2\theta\,d\theta$$

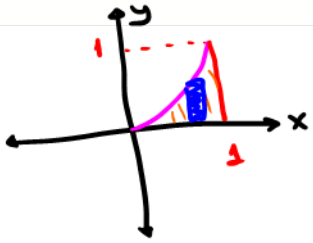
$$= \frac{81}{5}\cdot -\frac{\cos^3\theta}{3}\Big|_{\frac{\pi}{4}}^{\frac{5\pi}{4}} = \frac{81}{5}\left(\frac{1}{2\sqrt{2}}+\frac{1}{2\sqrt{2}}\right) = \frac{81}{5\sqrt{2}}$$

A şıkkı

11. $\int_0^1 \int_{\sqrt{y}}^1 \cos x^3 \, dx dy = ?$ → bu haliyle çözülemez! Sırasını değiştirmeliyim! dxdy → dydx olmalı.

a) $\frac{1}{3}\cos 1$ b) $\frac{1}{2}\sin 1$ c) $\frac{1}{3}\sin 1$ d) $\sin 1$ e) $\sin \frac{1}{3}$

$x = \sqrt{y}$, $x = 1$, $y = x^2$ $\quad y = 0$, $y = 1$



$$\int_0^1 \int_0^{x^2} \cos x^3 \, dy dx = \int_0^1 y \Big|_0^{x^2} \cdot \cos x^3 \, dx$$

$$= \int_0^1 \frac{3}{3} x^2 \cdot \cos x^3 \, dx = \frac{1}{3} \sin x^3 \Big|_0^1$$

$$= \frac{1}{3}\sin 1$$ C şıkkı

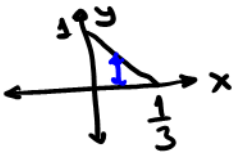
12. $x \geq 0$, $y \geq 0$, $z \geq 0$ olmak üzere koordinat düzlemleri ve $3x+y+z=1$ düzlemi ile sınırlı cismin hacmini veren integral aşağıdakilerden hangisidir?

a) $\int_0^{\frac{1}{3}} \int_0^{(1-3x)} (1-3x-y) \, dydx$ b) $\int_0^1 \int_0^{(6-3x)} (1+3x+y) \, dydx$ c) $\int_0^{\frac{1}{2}} \int_0^{(1-3x)} (-1-3x-y) \, dydx$

d) $\int_0^{\frac{1}{3}} \int_0^{(3x-1)} (-1+3x-y) \, dydx$ e) $\int_0^3 \int_0^{(1-3x)} (1-3x-y) \, dydx$

$$\iint_D (1-3x-y-0) \, dA$$

D: $3x+y+z=1$, $z=0$ $\Big]$ $3x+y=1$.



$$= \int_0^{1/3} \int_0^{1-3x} (1-3x-y) \, dydx$$ A şıkkı

13. D, $x+y=1$, $x+y=2$, $x=2y$ ve $x-2y=1$ egrileriyle sınırlı bölge olmak üzere $\iint_D (x-2y)\,dA$ integralinin değeri nedir?

a) $\frac{1}{2}$ b) 1 c) $\frac{1}{4}$ d) $\frac{3}{2}$ e) $\frac{1}{6}$

$x+y=u$
$x-2y=v$

$J = \frac{1}{\begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}} = \frac{1}{\begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix}} = -\frac{1}{3}$ $\quad dxdy = |J|.dudv = \frac{1}{3}dudv$

D': $u=1$, $u=2$, $v=0$, $v=1$

$$\int_1^2\int_0^1 v.\frac{1}{3}\,dvdu = \frac{1}{3}\int_1^2 \frac{v^2}{2}\Big|_0^1 du = \frac{1}{6}\int_1^2 du = \frac{1}{6}(2-1) = \frac{1}{6}$$

E şıkkı

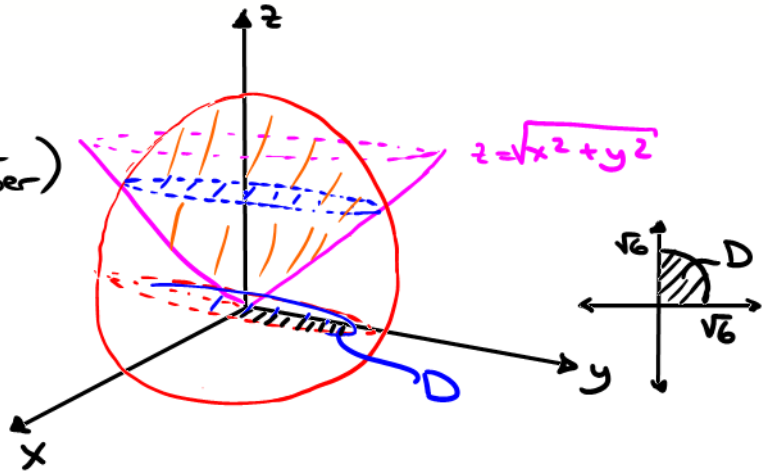
14. $x\geq 0$, $y\geq 0$, $z\geq 0$ olmak üzere üstten $x^2+y^2+z^2=12$ küresi ve alttan $z=\sqrt{x^2+y^2}$ konisi ile sınırlı cismin hacmini veren integral aşağıdakilerden hangisidir?

a) $\int_0^{\pi/3}\int_0^{\sqrt{6}}(\sqrt{12-r^2}-r)\,r\,dr\,d\theta$ b) $\int_0^{\pi/2}\int_0^{\sqrt{6}}(\sqrt{12-r^2}-r)\,r\,dr\,d\theta$ c) $\int_0^{\pi/2}\int_0^{6}(\sqrt{12-r^2}-r)\,r\,dr\,d\theta$

d) $\int_0^{\pi/2}\int_0^{\sqrt{6}}(\sqrt{6-r^2}-r)\,r\,dr\,d\theta$ e) $\int_0^{\pi/2}\int_0^{6}(\sqrt{6-r^2}-r)\,r\,dr\,d\theta$

$x^2+y^2+z^2=12$, $z=\sqrt{x^2+y^2}$, $x\geq 0$, $y\geq 0$, $z\geq 0$ $\Rightarrow$ $x^2+y^2=6$ (izdüşüm bölgesi)

D bölgesi (çeyrek çember)



$$\iint_D (\sqrt{12-x^2-y^2}-\sqrt{x^2+y^2})\,dA$$

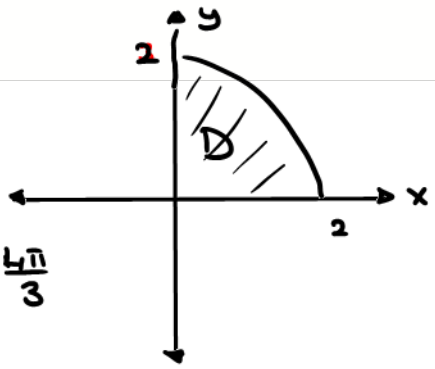
$x=r\cos\theta$, $y=r\sin\theta$ $\quad dxdy=r\,dr\,d\theta$

$$V=\int_0^{\frac{\pi}{2}}\int_0^{\sqrt{6}}(\sqrt{12-r^2}-r)\,r\,dr\,d\theta$$

B şıkkı

15. $f(x,y)=\sqrt{x^2+y^2}$ fonksiyonunun, $D=\{(x,y): 0\le x\le 2, 0\le y\le \sqrt{4-x^2}\}$ bölgesi üzerindeki ortalama değeri nedir?

a) $\frac{4}{3}$ b) $\frac{1}{3}$ c) $\frac{16}{3}$ d) $\frac{8}{3}$ e) $\frac{5}{3}$



$ortf = \frac{Hacim}{Alan}$ $\quad Alan = \frac{\pi r^2}{4} = \frac{\pi \cdot 2^2}{4} = \pi$

$$Hacim = \int_0^{\pi/2}\int_0^2 r\cdot r\, dr\, d\theta = \int_0^{\frac{\pi}{2}} \frac{r^3}{3}\Big|_0^2 d\theta = \frac{8}{3}\cdot\frac{\pi}{2} = \frac{4\pi}{3}$$

$$ortf = \frac{4\pi/3}{\pi} = \frac{4}{3}$$

A şıkkı

16. R bölgesi; $xy=1$, $xy=9$, $x=y$ ve $4x=y$ eğrilerinin sınırladığı bölge olsun. $\iint_R \left(\sqrt{\frac{y}{x}}+\sqrt{xy}\right) dA$ integralinin $u=\frac{y}{x}$ ve $v=xy$ dönüşümü altındaki değeri nedir?

a) $4+\frac{26}{3}\ln 2$ b) $9+\frac{26}{3}\ln 3$ c) $9+\frac{52}{3}\ln 2$ d) $8+\frac{52}{3}\ln 2$ e) $8+\frac{26}{3}\ln 3$

$xy=v$ $\quad \frac{y}{x}=u$ $\quad R'$: $u=1, u=4, v=1, v=9$

$$J = \frac{1}{\begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}} = \frac{1}{\begin{vmatrix} -y/x^2 & 1/x \\ y & x \end{vmatrix}} = \frac{1}{-2\frac{y}{x}} = \frac{-1}{2u}$$

$$dxdy = |J|\,du\,dv = \frac{1}{2u}\,du\,dv$$

$$\int_1^4\int_1^9 (\sqrt{u}+\sqrt{v})\frac{1}{2u}\,dv\,du = \int_1^4 \frac{1}{2u}\left(\sqrt{u}\cdot v + \frac{2}{3}v^{3/2}\right)\Big|_1^9 du = \int_1^4 \left(\frac{4}{\sqrt{u}} + (18-\frac{2}{3})\frac{1}{2u}\right) du$$

$$= \int_1^4 \left(\frac{4}{\sqrt{u}} + \frac{26}{3u}\right) du = 8\sqrt{u} + \frac{26}{3}\ln(u)\Big|_1^4 = 8(2-1) + \frac{26}{3}\ln 4$$

$$= 8 + \frac{52}{3}\ln 2$$

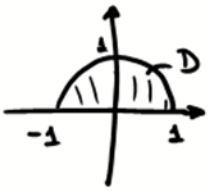
D şıkkı

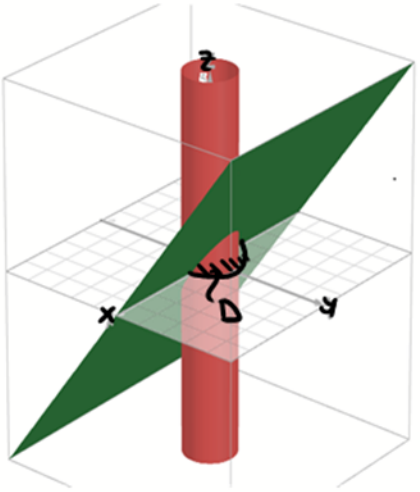
17. $x^2+y^2=1$ silindiri ile $z=0$ ve $z=y$ düzlemleri arasında kalan bölgenin hacmini veren integral aşağıdakilerden hangisidir?

a) $\int_0^{\pi}\int_0^{1} r^2\sin\theta\, dr d\theta$ b) $\int_0^{\pi}\int_0^{2} r^2\sin\theta\, dr d\theta$ c) $\int_0^{\pi}\int_{-1}^{1} r^2\cos\theta\, dr d\theta$

d) $\int_0^{\pi}\int_{-1}^{1} r^2\sin\theta\, dr d\theta$ e) $\int_0^{\pi}\int_0^{1} r^2\cos\theta\, dr d\theta$

$x^2+y^2=1$
$z=0$
$z=y$





$x=r\cos\theta$
$y=r\sin\theta$
$dx\,dy=r\,dr\,d\theta$

$$V=\iint_D (y-0)\,dA$$

$$V=\int_0^{\pi}\int_0^{1} r\sin\theta\cdot r\, dr d\theta$$

$$V=\int_0^{\pi}\int_0^{1} r^2\sin\theta\, dr d\theta$$

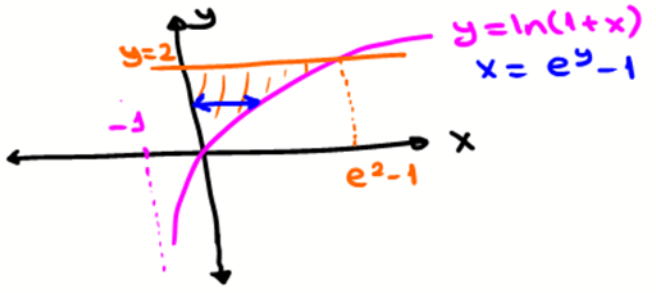
A şıkkı

18. $\int_0^{e^2-1}\int_{\ln(1+x)}^{2} f(x,y)\,dy dx$ integralinde integrasyon sırası değiştirildiğinde aşağıdaki integrallerden hangisi elde edilir?

a) $\int_0^{2}\int_0^{e^y-1} f(x,y)\,dx dy$ b) $\int_0^{1}\int_0^{e^y-1} f(x,y)\,dx dy$ c) $\int_{-2}^{2}\int_0^{e^y-2} f(x,y)\,dx dy$

d) $\int_0^{2}\int_1^{e^y-1} f(x,y)\,dx dy$ e) $\int_0^{2}\int_0^{e^y-2} f(x,y)\,dx dy$

$y=2$ $\quad x=0$
$y=\ln(1+x)$ $\quad x=e^2-1$



$$\int_{y=0}^{y=2}\int_0^{e^y-1} f(x,y)\,dx dy$$

// A şıkkı

19. $x^2+y^2=4y$ silindiri, $4z=x^2+y^2$ paraboloidi ve $z=0$ düzlemi arasında kalan cismin hacmini veren iki katlı kutupsal integral aşağıdakilerden hangisidir?

a) $\int_0^{\pi}\int_0^{4\sin\theta} \frac{r^3}{4}\,dr\,d\theta$ b) $\int_0^{\pi}\int_0^{4\sin\theta} r^3\,dr\,d\theta$ c) $\int_0^{\frac{\pi}{2}}\int_0^{4\cos\theta} \frac{r^2}{4}\,dr\,d\theta$

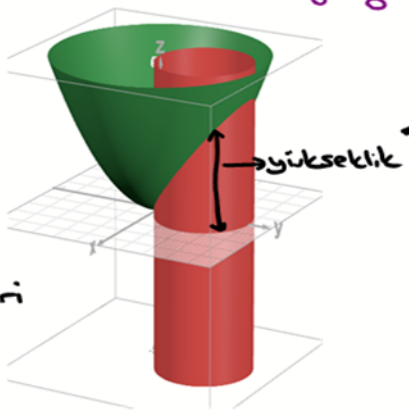
d) $\int_0^{\frac{\pi}{2}}\int_0^{2\sin\theta} \frac{r^2}{4}\,dr\,d\theta$ e) $\int_0^{\pi}\int_0^{4} \frac{r^3}{4}\,dr\,d\theta$

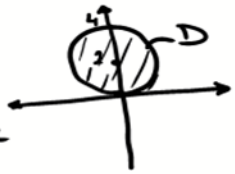
$$\begin{cases} x^2+y^2=4y \\ 4z=x^2+y^2 \\ z=0 \end{cases}$$

12 düzlem

$x^2+y^2=4y$ çemberi

$r=4\sin\theta$





$x=r\cos\theta$

$y=r\sin\theta$

$dx\,dy=r\,dr\,d\theta$

$$\iint_D \left(\frac{x^2+y^2}{4}-0\right) dA$$

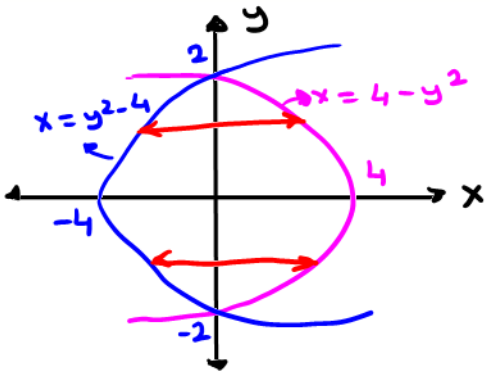
$$\int_0^{\pi}\int_0^{4\sin\theta} \frac{r^2}{4}\cdot r\,dr\,d\theta = \int_0^{\pi}\int_0^{4\sin\theta} \frac{r^3}{4}\,dr\,d\theta$$

A şıkkı

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20. $x=4-y^2$ ve $x=y^2-4$ eğrileri ile sınırlı düzlem bölgenin alanını veren integral aşağıdakilerden hangisidir?

a) $\int_{-2}^{2}\int_{4-y^2}^{y^2-4} dx\,dy$ b) $\int_{-2}^{2}\int_{4-y^2}^{y^2-4} dy\,dx$ c) $\int_{0}^{4}\int_{4-y^2}^{y^2-4} dy\,dx$ d) $\int_{-2}^{2}\int_{y^2-4}^{4-y^2} dy\,dx$ e) $\int_{-2}^{2}\int_{y^2-4}^{4-y^2} dx\,dy$



↓ şıklardaki integraller tek parçalı o yüzden bölge dxdy ile hesaplanmış

$$A=\int_{y=-2}^{y=2}\int_{y^2-4}^{4-y^2} dx\,dy$$

E şıkkı

21. D bölgesi; $x-y=2$, $x-y=4$, $x+y=1$ ve $x+y=2$ doğruları ile sınırlı bölge olsun. $\iint_D (x^2-y^2)\,dxdy$ integralinin sınırlarını $u=x-y$, $v=x+y$ değişken dönüşümlerini kullanarak yazdığımızda elde edeceğimiz integral aşağıdakilerden hangisidir?

a) $\int_1^2 \int_2^4 \frac{uv}{2}\,dudv$ b) $\int_2^4 \int_1^2 uv\,dudv$ c) $\int_1^2 \int_2^4 uv\,dudv$ d) $\int_2^4 \int_1^2 \frac{uv}{2}\,dudv$

e) $\int_2^4 \int_1^2 u^2v^2\,dudv$

$u=x-y$ $\quad D'$: $u=2, u=4$

$v=x+y$ $\quad\quad v=1, v=2$

$$J = \frac{1}{\begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}} = \frac{1}{\begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix}} = \frac{1}{2}$$

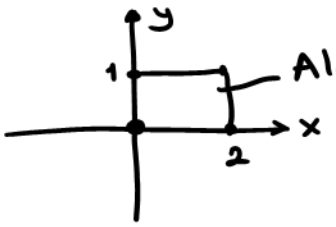
$$\iint_D (x^2-y^2)\,dxdy = \iint_D (x-y)(x+y)\,dxdy$$

$$= \int_1^2 \int_2^4 \frac{u.v}{2}\,dudv$$

A şıkkı

22. Köşeleri $(0,0)$, $(0,1)$, $(2,0)$ ve $(2,1)$ noktalarında bulunan dikdörtgensel bir bölgede $f(x,y)=xy+2x$ fonksiyonunun ortalama değeri aşağıdakilerden hangisidir?

a) 3 b) 2 c) $\frac{5}{2}$ d) $\frac{7}{2}$ e) $\frac{9}{2}$



Alan $= 2 \cdot 1 = 2$

$$ort f = \frac{Hacim}{Alan}$$

$$Hacim = \int_0^2 \int_0^1 (xy+2x)\,dy\,dx = \int_0^2 x\frac{y^2}{2} + 2xy \Big|_0^1 dx$$

$$= \int_0^2 \frac{5}{2}x\,dx = \frac{5}{4}x^2\Big|_0^2 = 5$$

$$ort f = \frac{5}{2}$$

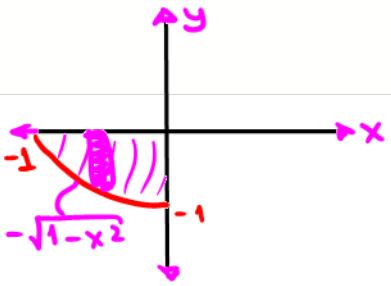
C şıkkı

23. Aşağıdaki integrallerden hangisi $\int_{-1}^{0}\int_{-\sqrt{1-x^2}}^{0}(x^2+y^2)\,dydx$ integralinin kutupsal koordinatlarda yazılmış halidir?

a) $\int_{\frac{3\pi}{2}}^{2\pi}\int_{0}^{1} r^3\,dr\,d\theta$ b) $\int_{\pi}^{\frac{3\pi}{2}}\int_{0}^{1} r^3\,dr\,d\theta$ c) $\int_{\frac{\pi}{2}}^{\pi}\int_{0}^{1} r^3\,dr\,d\theta$ d) $\int_{0}^{\frac{\pi}{2}}\int_{0}^{1} r^3\,dr\,d\theta$

e) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\int_{0}^{1} r^3\,dr\,d\theta$

$$\int_{-1}^{0}\int_{-\sqrt{1-x^2}}^{0}(x^2+y^2)\,dydx$$



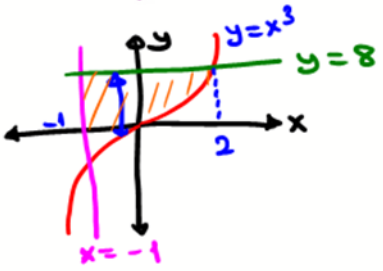
$x=r\cos\theta$
$y=r\sin\theta$
$dxdy=r\,dr\,d\theta$

$$=\int_{\pi}^{\frac{3\pi}{2}}\int_{0}^{1} r^2.r\,dr\,d\theta=\int_{\pi}^{\frac{3\pi}{2}}\int_{0}^{1} r^3\,dr\,d\theta$$

B şıkkı

24. D bölgesi ; $y=x^3$ eğrisi ile $y=8$ ve $x=-1$ doğruları ile sınırlı bölge olmak üzere, D bölgesinin alanını veren iki katlı integral aşağıdakilerden hangisidir?

a) $\int_{-1}^{2}\int_{x^3}^{-1} dydx$ b) $\int_{-1}^{2}\int_{-8}^{x^3} dydx$ c) $\int_{-1}^{2}\int_{\sqrt[3]{y}}^{8} dydx$ d) $\int_{-1}^{2}\int_{x^3}^{8} dydx$ e) $\int_{-1}^{2}\int_{x^3}^{0} dydx$



$$A=\int_{x=-1}^{x=2}\int_{x^3}^{8} dydx$$

D şıkkı

25. $\int_0^{\frac{2}{3}} \int_{2x}^{2-x} f(x,y)\,dydx$ integralini integrasyon sırasını değiştirerek yeniden yazdığımızda elde edilen integral aşağıdakilerden hangisidir?

a) $\int_0^{\frac{2}{3}} \int_0^{\frac{y}{2}} f(x,y)\,dxdy + \int_{\frac{2}{3}}^{2} \int_{\frac{y}{2}}^{2-y} f(x,y)\,dxdy$

b) $\int_{2x}^{2-x} \int_0^{\frac{2}{3}} dxdy$

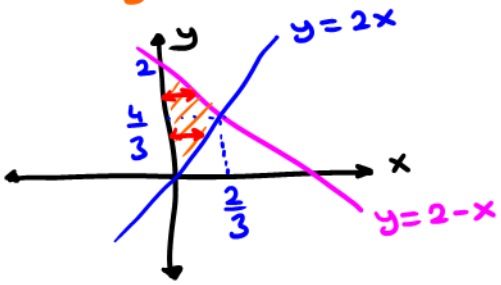
c) $\int_0^{\frac{4}{3}} \int_0^{\frac{y}{2}} f(x,y)\,dxdy + \int_{\frac{4}{3}}^{2} \int_{\frac{y}{2}}^{2-y} f(x,y)\,dxdy$

d) $\int_0^{\frac{2}{3}} \int_0^{\frac{y}{2}} f(x,y)\,dxdy + \int_0^{\frac{4}{3}} \int_0^{2-y} f(x,y)\,dxdy$

e) $\int_0^{\frac{4}{3}} \int_0^{\frac{y}{2}} f(x,y)\,dxdy + \int_{\frac{4}{3}}^{2} \int_0^{2-y} f(x,y)\,dxdy$

$x=0 \quad y=2x$

$x=\frac{2}{3} \quad y=2-x$



$$\int_{y=0}^{\frac{4}{3}} \int_0^{\frac{y}{2}} f(x,y)\,dxdy + \int_{\frac{4}{3}}^{2} \int_0^{2-y} f(x,y)\,dxdy$$

E şıkkı

26. R bölgesi; $x=\sqrt{y}$ eğrisi ve $x+y=2$; $x=0$ doğruları ile sınırlanmış bölge olsun. R bölgesinin alanını veren integral aşağıdakilerden hangisidir?

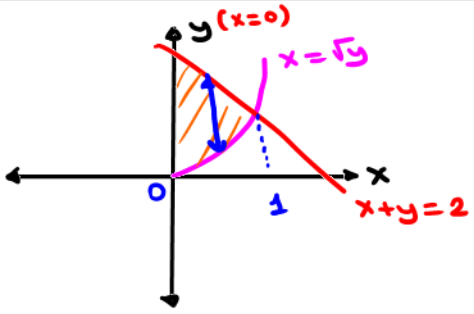
a) $\int_0^1 \int_{\sqrt{x}}^{2-x} dxdy$

b) $\int_0^1 \int_{y^2}^{2-y} dydx$

c) $\int_0^1 \int_{x^2}^{2-x} dydx$

d) $\int_0^1 \int_{\sqrt{y}}^{2-y} dxdy$

e) $\int_0^1 \int_{2-x}^{x^2} dydx$



$$A = \int_{x=0}^{x=1} \int_{x^2}^{2-x} dydx$$

C şıkkı

27. $\int_0^2 \int_0^{\sqrt{2y-y^2}} \frac{xy}{x^2+y^2}\, dx\,dy$ iki katlı integrali kutupsal koordinatlarda yeniden yazıldığında aşağıdaki integrallerden hangisi elde edilir?

a) $\int_0^{\frac{\pi}{2}} \int_0^{2\sin\theta} \sin\theta.\cos\theta\, r\,dr\,d\theta$ b) $\int_0^{\frac{\pi}{2}} \int_0^{2\cos\theta} \sin\theta\cos\theta\, r\,dr\,d\theta$ c) $\int_0^{\frac{\pi}{2}} \int_0^{\sin\theta} \tan\theta\, r\,dr\,d\theta$

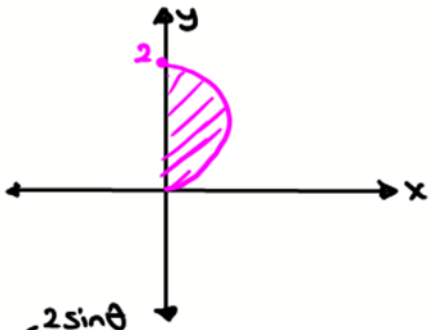
d) $\int_0^{\frac{\pi}{2}} \int_0^{2\sin\theta} \sin\theta.\cos\theta\, dr\,d\theta$ e) $\int_0^{2\pi} \int_{\cos\theta}^{\sin\theta} \sin\theta.\cos\theta\, r\,dr\,d\theta$

$y=0$, $y=2$

$x=0$, $x=\sqrt{2y-y^2}$

$x^2+y^2=2y$

$r=2\sin\theta$



$x = r\cos\theta$

$y = r\sin\theta$

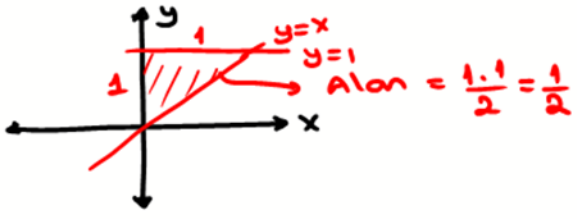
$dx\,dy = r\,dr\,d\theta$

$$\iint_D \frac{xy}{x+y^2}\, dA = \int_0^{\frac{\pi}{2}} \int_0^{2\sin\theta} \frac{r\cos\theta.r\sin\theta}{r2}.r\,dr\,d\theta = \int_0^{\frac{\pi}{2}} \int_0^{2\sin\theta} r\sin\theta.\cos\theta\, dr\,d\theta$$

A şıkkı

28. R bölgesi; $x=0$, $y=1$ ve $y=x$ doğruları ile kapalı bölge olsun. R bölgesinde $f(x,y)=xy$ fonksiyonunun ortalama değeri nedir?

a) $\frac{1}{4}$ b) $\frac{2}{15}$ c) $\frac{1}{9}$ d) $\frac{27}{2}$ e) $\frac{1}{5}$



Alan $= \frac{1.1}{2} = \frac{1}{2}$

$ortf = \frac{Hacim}{Alan}$

$ortf = \frac{1/8}{1/2} = \frac{1}{4}$

A şıkkı

$$Hacim = \int_0^1 \int_x^1 xy\, dy\,dx$$

$$= \int_0^1 x\,\frac{y^2}{2}\Big|_x^1 dx$$

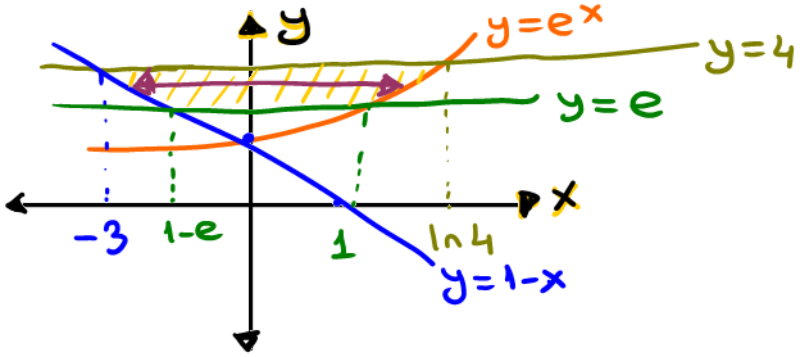
$$= \int_0^1 \left[\frac{x}{2} - \frac{x^3}{2}\right] dx = \frac{x^2}{4} - \frac{x^4}{8}\Big|_0^1$$

$$= \frac{1}{4} - \frac{1}{8} = \frac{1}{8}$$

29. $\int_{-3}^{1-e}\int_{1-x}^{4} f(x,y)\,dydx + \int_{1-e}^{1}\int_{e}^{4} f(x,y)\,dydx + \int_{1}^{\ln 4}\int_{e^x}^{4} f(x,y)\,dydx = ?$

a) $\int_{e}^{4}\int_{1-y}^{\ln y} f(x,y)\,dxdy$ b) $\int_{e}^{4}\int_{1+y}^{\ln y} f(x,y)\,dxdy$ c) $\int_{0}^{1}\int_{e}^{e^y} f(x,y)\,dxdy$

d) $\int_{-3}^{1}\int_{1+y}^{e^y} f(x,y)\,dxdy$ e) $\int_{-3}^{1}\int_{e^y}^{4} f(x,y)\,dxdy$



$\int_{y=e}^{y=4}\int_{x=1-y}^{x=\ln y} f(x,y)\,dxdy$

A şıkkı

30. f, diferansiyellenebilir bir çift fonksiyon olmak üzere, $\int_0^1 f(y)\,dy = 2$ olsun. $\int_{-1}^{1}\int_{0}^{1} f(y)\,xe^{\frac{x^2}{2}}\,dxdy = ?$

a) $2\sqrt{e}-2$ b) $2e-1$ c) $4\sqrt{e}-4$ d) $4e-1$ e) 0

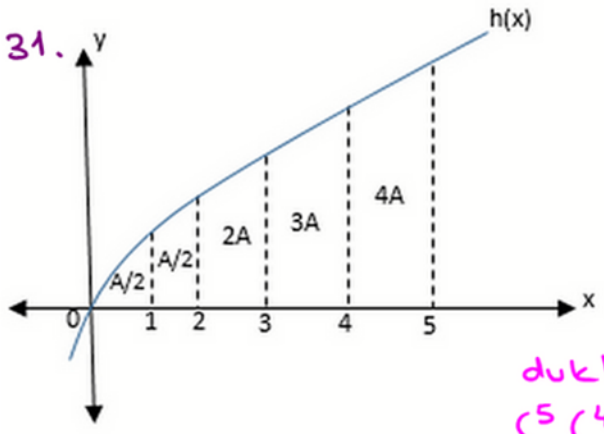
$$\int_{-1}^{1}\int_{0}^{1} f(y)\,xe^{x^2/2}\,dxdy = \int_{-1}^{1} e^{x^2/2}\Big|_0^1 f(y)\,dy$$

$$= \int_{-1}^{1}(\sqrt{e}-1)\,f(y)\,dy = (\sqrt{e}-1)\,2\cdot\underbrace{\int_0^1 f(y)\,dx}_{2} = 4\sqrt{e}-4$$

$f(y)$: çift fonk.

C şıkkı

31.



$f(x,y)$ ve $h(x)$ diferansiyellenebilir fonksiyonlar olmak üzere $\int_1^5 f(x,y)\,dy = h(x)$ ve $\int_0^4 h(x)\,dx = 12$ olsun. Yandaki grafikte, $A/2$, $2A$, $3A$, $4A$ bulunduklari bölgenin alanlarini temsil etmektedir.

$\int_1^5 \int_2^4 f(x,y)\,dx\,dy = ?$

a) 9 b) 10 c) 18 d) 19 e) 20

$$\int_0^4 h(x)\,dx = 3A + 2A + \frac{A}{2} + \frac{A}{2} = 6A = 12 \Rightarrow \underline{A = 2br^2}$$

$$\int_1^5 \int_2^4 f(x,y)\,dx\,dy = \int_2^4 \underbrace{\int_1^5 f(x,y)\,dy}_{h(x)}\,dx = \int_2^4 h(x)\,dx = 5A$$

$$= 5 \cdot 2 = \underline{10br^2}$$

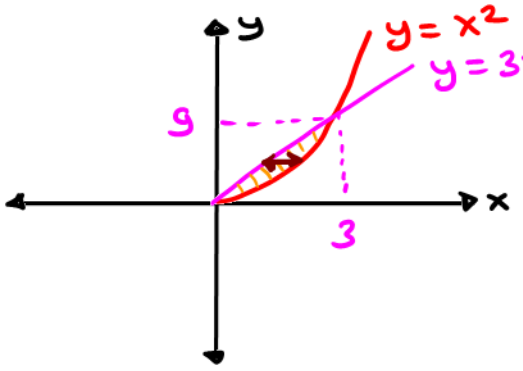
B şıkkı

32. $\int_0^3 \int_{x^2}^{3x} f(x,y)\,dy\,dx$ integralinde integrasyon sırası değiştirildiğinde elde edilen integral aşağıdakilerden hangisidir?

a) $\int_0^3 \int_y^{\sqrt{y}} f(x,y)\,dx\,dy$ b) $\int_0^{16} \int_{\frac{y}{4}}^{\sqrt{y}} f(x,y)\,dx\,dy$ c) $\int_0^{25} \int_{\frac{y}{5}}^{\sqrt{y}} f(x,y)\,dx\,dy$

d) $\int_0^4 \int_{\frac{y}{2}}^{\sqrt{y}} f(x,y)\,dx\,dy$ e) $\int_0^9 \int_{\frac{y}{3}}^{\sqrt{y}} f(x,y)\,dx\,dy$

$x = 0$ $y = x^2$
$x = 3$ $y = 3x$



$$\int_{y=0}^{y=9} \int_{\frac{y}{3}}^{\sqrt{y}} f(x,y)\,dx\,dy$$

E şıkkı

33. D bölgesi; $xy=1$, $xy=3$ eğrileri ve $y=2$, $y=6$ doğruları ile sınırlı bölge olsun. $\iint_D xy^3\,dA$ integralinde, $x=\frac{v}{6u}$, $y=2u$ değişken dönüşümü kullanıldığında aşağıdaki integrallerden hangisi elde edilir?

a) $\frac{4}{9}\int_3^9\int_1^3 uv\,du\,dv$ b) $\frac{2}{9}\int_6^9\int_1^2 uv\,du\,dv$ c) $\frac{2}{3}\int_3^{12}\int_1^2 uv\,du\,dv$

d) $\frac{2}{9}\int_3^9\int_1^3 uv\,du\,dv$ e) $\frac{4}{9}\int_2^4\int_1^3 uv\,du\,dv$

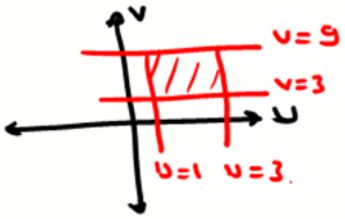
$x\cdot y = \underbrace{\frac{v}{6u}}_{x}\cdot\underbrace{2u}_{y} = \frac{v}{3}$

$xy=1 \to v=3$

$xy=3 \to v=9$

$y=2 \to u=1$

$y=6 \to u=3$



$$J=\begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = \begin{vmatrix} \frac{-v}{6u^2} & \frac{1}{6u} \\ 2 & 0 \end{vmatrix} = -\frac{1}{3u}$$

$dx\,dy = |J|\,du\,dv = \frac{1}{3u}\,du\,dv$

$$\iint_D xy^3\,dA = \int_3^9\int_1^3 \frac{v}{3}\cdot 4u^2\cdot\frac{1}{3u}\,du\,dv = \int_3^9\int_1^3 \frac{4}{9}\,u.v\,du\,dv$$

A şıkkı